

Comments on
*Fully accounted NLO electromagnetic radiative corrections
to the width of J/ψ decay*

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Equation, table and reference numbers refer to the current draft. “The note” is *Radiative Corrections to the J/ψ photoproduction reaction*; Ref. [10] of the draft is Ehlotzky & Mitter, *Nuovo Cimento* **55A** (1968) 181, abbreviated E&M below.

Sections 1–4 are small textual points. Sections 5–8 are substantive.

1 Eq. (1) and Eq. (17): replace E_V by m_V

Both read $1/(2E_V)$. The calculation is done in the J/ψ rest frame, where $E_V = m_V$ identically, and the text already says so. Writing m_V removes a symbol that never takes any other value and makes the normalisation of Eq. (1) directly comparable with the standard two-body form.

2 Eq. (6): use the factorised form, and add the ultrarelativistic limit

Eq. (6) is correct — algebraically identical to Eq. (7) of the note (verified symbolically; the difference is exactly zero). But the factorised form is easier to read and to check dimensionally, since the mass enters only through dimensionless ratios:

$$\Gamma_B = \frac{4\pi\alpha^2}{f_V^2} \frac{m_V}{3} \left(1 + \frac{2m_\ell^2}{m_V^2}\right) \sqrt{1 - \frac{4m_\ell^2}{m_V^2}}.$$

Please also add the ultrarelativistic limit, Eq. (8) of the note, which is the form actually used to fix f_V from the measured leptonic width:

$$\Gamma_B = \frac{4\pi\alpha^2}{f_V^2} \frac{m_V}{3}.$$

Worth stating explicitly: the correction connecting the two is *second* order in the mass ratio. With $x = m_\ell^2/m_V^2$,

$$(1 + 2x)\sqrt{1 - 4x} = 1 - 6x^2 + O(x^3),$$

so the linear term cancels. This is why the massless limit is so accurate: 4×10^{-15} for $J/\psi \rightarrow e^+e^-$, 8×10^{-6} for $J/\psi \rightarrow \mu^+\mu^-$. It reaches 2×10^{-3} only for $\rho \rightarrow \mu^+\mu^-$.

3 Eq. (4): the polarization sum

As printed, $\sum_{\text{pol}} \varepsilon_\alpha(p_V) \varepsilon_\beta(p_V) = -g_{\alpha\beta}$ is the polarization sum for a *massless* vector. For a massive vector meson it is $-g_{\alpha\beta} + p_\alpha p_\beta / m_V^2$. The extra term does drop here, because it contracts the lepton current with $p_V = p_1 + p_2$ and $\bar{u}(p_1)(\not{p}_1 + \not{p}_2)v(p_2) = 0$ by the Dirac equations — so Eq. (5) and everything after it are correct. But the identity as written is not true on its own, and the surrounding text calls it “the photon” while the sum is over the three J/ψ polarizations. A clause such as “the $p_\alpha p_\beta$ term is dropped by current conservation” would fix it.

4 Reference [9]

[9] P. Chatagnon, e-Print: 2602.22128 [hep-ex] (2026) is now published. Please replace with [9] P. Chatagnon, Phys. Rev. C 113 (2026) 6, 065203.

5 Please add the closed-form total correction, for both channels

The paper gives δ_{self} (8), δ_{vert} (12), δ_{soft} (13), and δ_{hard} only as an unevaluated integral (14), then the exact result (28). There is no compact closed form for the total correction — the quantity an experiment applies, and what a reader comes to this paper looking for. These are Eqs. (19) and (20) of the note, i.e. Eq. (26) of Ref. [10] with the missing term restored.

Electrons, $J/\psi \rightarrow e^+e^-$:

$$\delta(m_V, k_{\text{max}}) = -\frac{4\alpha}{\pi} \left[\left(\frac{1}{2} - \ln \frac{m_V}{m_e} \right) \left(\ln r_{\text{max}} + \frac{1}{3} \right) + \frac{1}{2} \left(L_2(r_{\text{max}}) - \frac{\pi^2}{6} \right) + \frac{17}{72} + R(r_{\text{max}}) + \frac{r_{\text{max}}^2}{16} - \frac{3}{8} r_{\text{max}} \right]$$

Muons, $J/\psi \rightarrow \mu^+\mu^-$:

$$\delta(m_V, k_{\text{max}}) = -\frac{4\alpha}{\pi} \left[\left(\frac{1}{2} - \ln \frac{m_V}{m_\mu} \right) \left(\ln r_{\text{max}} + \frac{1}{3} \right) + \frac{1}{2} \left(L_2(r_{\text{max}}) - \frac{\pi^2}{6} \right) + \frac{17}{72} \right. \\ \left. + \left(\frac{5}{18} - \frac{1}{3} \ln \frac{m_V}{m_e} \right) + R(r_{\text{max}}) + \frac{r_{\text{max}}^2}{16} - \frac{3}{8} r_{\text{max}} \right]$$

with, in each case,

$$r_{\text{max}} = \frac{2k_{\text{max}}}{m_V}, \quad R(r_{\text{max}}) = \left(r_{\text{max}} - \frac{r_{\text{max}}^2}{4} - \frac{3}{4} \right) \left[\ln \left(\frac{m_V}{m_\ell} \sqrt{1 - r_{\text{max}}} \right) - \frac{1}{2} \right], \quad L_2(x) = - \int_0^x \frac{\ln(1-t)}{t} dt,$$

m_ℓ being the lepton of the channel concerned. The extra term in the muon expression, $\frac{5}{18} - \frac{1}{3} \ln \frac{m_V}{m_e}$, is the electron-loop vacuum polarization; it is exactly $-\frac{\pi}{4\alpha}$ times δ_{self} of Eq. (8) evaluated with m_e , and it correctly keeps m_e , not m_μ . Here r_{max} is E&M’s variable, k_{max}/E with $E = m_V/2$; it is identical to the $r_{\text{cut}} = \omega_{\text{cut}}/E_1$ already defined in this paper. The last two terms of each bracket are the ones absent from [10].

Where it belongs

Section IV, immediately after Eq. (14) — the closed form *is* the evaluation of that integral combined with Eqs. (8), (12) and (13).

- Section IV opens by promising “compact expressions for all contributions to decay width: both for virtual and real bremsstrahlung ones.” Compact expressions follow for every individual contribution, but not for the sum, which is the only physical combination.
- Eq. (16), $\delta_{FSIC} = 3\alpha/4\pi$, then becomes an immediate corollary: it is the $r_{max} \rightarrow 1$ limit of the closed form, less δ_{self} .
- Figure 4 plots exactly this quantity. At present the reader is shown the curve without being given the function.

Section IV also states that the URA is “applicable for $\ell = e$ ” and that the mass-dependent treatment is “crucial for ... the muon channel”. We would therefore present the electron expression as the working formula of Section IV, and the muon one alongside with the $(m_\mu/E)^2$ error flagged and Eq. (28) named as the fallback.

Status of the approximation

These expressions are *not* exact in the lepton mass, and not the $m_\ell = 0$ limit either: the mass is kept in the logarithms, while terms suppressed by $(m_\ell/E)^2$, $E = m_V/2$, are dropped. Ref. [10] states this just above its Eq. (22) (“... neglecting terms of order $(m/E)^2$ ”).

	$(m_\ell/E)^2$	δ_{self} : Eq. (8)–Eq. (9)	total at $r_{max} = 1$: this form – Eq. (28)
e^+e^-	1.1×10^{-7}	$+3 \times 10^{-10}$	$+1 \times 10^{-7}$
$\mu^+\mu^-$	4.7×10^{-3}	$+1.1 \times 10^{-5}$	-7×10^{-5}

So for electrons the closed form is exact at any precision an experiment will reach; for muons it is good to $\sim 10^{-4}$ absolute. Presenting it alongside Eq. (28) lets a reader use the simple expression where it suffices and the exact one where it does not.

Checks these expressions pass: they reproduce Table I of the draft to all six digits at every r_{cut} , in both channels, and are independent of ω by construction; at $r_{max} = 1$ they give exactly $\delta_{self} + 3\alpha/4\pi$; and for $J/\psi \rightarrow e^+e^-$ at $r_{max} = 1$ the value is 0.026135, matching the Bardin–Shumeiko result of Table III.

6 The missing term originates in Ref. [10]

E&M Eq. (23), the differential spectrum, is correct and agrees with Eq. (14) of the draft. The term was lost when that spectrum was integrated analytically in 1968, so Eqs. (24) and (26) of [10] are both incomplete. The note transcribed those equations faithfully and inherited the omission; it has since been corrected.

Where to record this: in Section IV, immediately after the closed-form expressions of Section 5 above — right where the new formulas will appear, following Eq. (14). A single paragraph there attributes the formula, records the correction, and explains the numerical comparison with [10] all at once:

The expressions above are Eq. (26) of Ref. [10] with the term $(\alpha/\pi)(3r_{max}/2 - r_{max}^2/4)$ restored. The differential spectrum, Eq. (23) of [10], is correct and agrees with our Eq. (14); the term was lost in its analytic integration, so that Eq. (24), Eq. (26) and Table I of [10] are all low by that amount — $5\alpha/4\pi = 0.0029$ at $r_{max} = 1$, and below the precision quoted in [10] for $r_{max} \lesssim 0.1$.

7 Table I of Ref. [10] was computed with the incomplete formula

This matters for the validation in Section IV, because Table I of [10] is not a correct benchmark: it was computed from E&M’s own Eq. (26) and carries the same omission. The evidence is not one line but the whole table — the δ column at $r_{max} = 1.00$, all fourteen mass rows, both channels, 28 entries:

agreement with the δ printed in Table I of [10]	electrons	muons
E&M Eq. (26) <i>as published</i> (term absent)	12/14 rows exact	12/14 rows exact
E&M Eq. (26) <i>corrected</i>	0/14	0/14

The two misses in each channel are last-digit rounding, not disagreement; the corrected formula misses every row by $\simeq 0.003$. The agreement reported in the draft at $\omega/m_e = 0.01$ therefore holds only in the small- r_{cut} rows, where the missing term falls below the three decimals quoted in [10].

Table I of the draft is itself correct — it follows from Eqs. (8), (12), (13), (14), which are all sound. It is only the benchmark that is at fault. Two ways to handle this, not exclusive:

(1) State it, using the paragraph of Section 6 above. The sentence claiming the comparison “succeeds” should become a pointer to it, since as written it asserts agreement with a table that is in error.

(2) Recompute Table I at $m_V = M(J/\psi)$. Since [10] can no longer serve as the validation target, a table at the physical mass is more useful: it demonstrates the ω -independence just as well, and the numbers are ones a J/ψ experiment can use. For $m_V = 3.0969$ GeV, in the format of the draft:

r_{cut}	$\lg(\omega/m_\ell)$	e^-e^+			$\mu^-\mu^+$		
		δ_R	δ_H	δ	δ_R	δ_H	δ
0.10	−6	−1.577194	1.481750	−0.095444	−0.382419	0.376596	−0.005823
	−4	−1.225925	1.130481	−0.095444	−0.259274	0.253452	−0.005823
	−2	−0.874655	0.779212	−0.095443	−0.136130	0.130328	−0.005801
0.25	−6	−1.577194	1.540558	−0.036636	−0.382419	0.396794	+0.014375
	−4	−1.225925	1.189288	−0.036636	−0.259274	0.273649	+0.014375
	−2	−0.874655	0.838019	−0.036636	−0.136130	0.150526	+0.014396
0.50	−6	−1.577194	1.576919	−0.000275	−0.382419	0.408881	+0.026462
	−4	−1.225925	1.225649	−0.000275	−0.259274	0.285736	+0.026462
	−2	−0.874655	0.874380	−0.000275	−0.136130	0.162613	+0.026483
1.00	−6	−1.577194	1.603329	+0.026135	−0.382419	0.416436	+0.034017
	−4	−1.225925	1.252060	+0.026135	−0.259274	0.293291	+0.034017
	−2	−0.874655	0.900791	+0.026136	−0.136130	0.170168	+0.034038

δ_R for the muon channel includes both self-energy loops. The $r_{cut} = 1.00$ electron entry, 0.026135,

equals $\delta_{self} + 3\alpha/4\pi$ and matches Table III.

8 The upper limit of r_{cut} is not 1

The text before Eq. (16) says the cut energy takes its maximum “at (15) or $r_{cut} = r_{max} = 1$ ”, and Table III is captioned ($r_{max} = 1$). That is inconsistent with Eq. (15). With $r_{cut} = 2\omega_{cut}/m_V$ and $\omega_{max} = m_V/2 - 2m_\ell^2/m_V$,

$$r_{cut}^{\max} = \frac{2\omega_{max}}{m_V} = 1 - \frac{4m_\ell^2}{m_V^2} = \beta^2,$$

which is 0.99535 for muons, not 1. ($1 - 1.1 \times 10^{-7}$ for electrons.)

This is not only notational. Eq. (29) defines $\rho_1(\omega_{cut}) = \sqrt{m_V(m_V - 2\omega_{cut})} + \sqrt{m_V(m_V - 2\omega_{cut}) - 4m_\ell^2}$. Setting $\omega_{cut} = m_V/2$, i.e. literally $r_{cut} = 1$, makes the second radicand $-4m_\ell^2 < 0$ and ρ_1 imaginary. At the true endpoint the radicand vanishes exactly and $\rho_1(\omega_{max}) = 2m_\ell$, the correct finite limit; Eq. (28) is well defined up to ω_{max} and no further.

We take it that Table III was evaluated at $\omega_{cut} = \omega_{max}$ and only the label is wrong. We suggest replacing “ $r_{cut} = r_{max} = 1$ ” and the Table III caption by “ $\omega_{cut} = \omega_{max}$ ” or “ $r_{cut} = \beta^2$ ”. The axis of Fig. 4 is already correctly drawn as $\omega_{cut}/\omega_{max}$.

9 Status of the note

The note has been corrected: the missing term in its Eqs. (14), (15), (19), (20); the leading logarithm of its Eq. (20), which had m_e where m_μ belongs; the phase-space factor of its Eq. (6); and the maximum photon energy, which now carries the exact endpoint and the bound $r_{max} \leq \beta^2$. Its Fig. 8 has been redrawn against r_{max} over the full physical range, so it now coincides with Fig. 4 of this draft — the two abscissae are related by $r_{max} = (\omega_{cut}/\omega_{max})\beta^2$, and the curves agree to machine precision in both channels. A paragraph giving the correspondence between the variables of the two papers ($\varepsilon \leftrightarrow \omega$, $k_{max} \leftrightarrow \omega_{cut}$, $r_{max} \leftrightarrow r_{cut}$, $\ln(m_V/m_\ell) = L/2$) has been added.